



THANOS: A Predictive Model of Electoral Campaigns Using Twitter Data and Opinion Polls

Dhrubajyoti Ghosh, William A. Boettcher, Rob Johnston & Soumendra Lahiri

To cite this article: Dhrubajyoti Ghosh, William A. Boettcher, Rob Johnston & Soumendra Lahiri (2025) THANOS: A Predictive Model of Electoral Campaigns Using Twitter Data and Opinion Polls, Data Science in Science, 4:1, 2484180, DOI: [10.1080/26941899.2025.2484180](https://doi.org/10.1080/26941899.2025.2484180)

To link to this article: <https://doi.org/10.1080/26941899.2025.2484180>



© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC



Published online: 01 Apr 2025.



Submit your article to this journal [↗](#)



Article views: 2218



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 1 View citing articles [↗](#)

THANOS: A Predictive Model of Electoral Campaigns Using Twitter Data and Opinion Polls

Dhrubajyoti Ghosh^a , William A. Boettcher^b, Rob Johnston^b, and Soumendra Lahiri^c

^aDepartment of Biostatistics and Bioinformatics, Duke University, Durham, NC, USA; ^bDepartment of Political Science, North Carolina State University, Raleigh, NC, USA; ^cDepartment of Mathematics & Statistics, Washington University in St. Louis, St. Louis, MO, USA

ABSTRACT

The influence and impact of social media campaigns on democratic elections is a critical area of research in modern big-data analytics. While the efficacy of using social media data for forecasting election results has been debated in political and social science literature, this paper introduces a novel modeling approach that combines public opinion polls with Twitter data, incorporating key network structure features of Twitter to enhance prediction accuracy. We developed two models: the Twitter Hashtag based Opinion Survey (THOS) model, which uses hashtag frequency, and the Twitter Hashtag and Network-based Opinion Survey (THANOS) model, which includes network centrality measures. Applying these models to Ireland's 36th amendment referendum and the 2018 US Senate elections yielded promising results. The THOS model effectively predicted outcomes in races with clear frontrunners, while the THANOS model excelled in closely contested races by leveraging network dynamics. These findings underscore the potential of integrating social media data with traditional polls to improve electoral forecasts, demonstrating the robust capabilities of the THOS and THANOS models in providing accurate predictions based on the interplay of public opinion and social media activity.

ARTICLE HISTORY

Received 22 October 2024
Revised 17 February 2025
Accepted 19 March 2025

KEYWORDS

Election forecasting; social media analytics; Twitter data; public opinion polls; network analysis; predictive modelling

1. Introduction

The rise of social media has been one of the most transformative developments of the twenty first century, reshaping communication and information sharing. Platforms like Twitter, Facebook, and MySpace have provided individuals with unprecedented opportunities to express their views and engage with broader communities at local, state, national, and international levels. This shift has had profound implications for marketing, politics, and social interactions. Political campaigns, in particular, have increasingly relied on social media to engage citizens and disseminate their messages. The US presidential campaigns of Barack Obama utilized social media extensively, and subsequent campaigns, including those of Donald Trump in 2016 and 2020, employed sophisticated social media strategies to target specific audiences. Consequently, social media has become a valuable resource for analyzing political and societal trends, including electoral outcomes.

Several studies have explored the potential of social media data to predict electoral outcomes, resulting in two distinct schools of thought: those who believe Twitter data can be effectively used for election prediction and those who argue it cannot. The first school of thought, advocating for the use of Twitter data, began with Tumasjan et al. (2010), who analyzed the sentiment and volume of tweets to

forecast the 2009 German federal election results, suggesting a positive correlation between social media activity and electoral success. O'Connor et al. (2010) analyzed Twitter data to predict the 2008 US Presidential Election, highlighting the feasibility of using tweet sentiment as an indicator of public opinion. Bermingham and Smeaton (2011) used sentiment analysis to forecast the 2011 Irish General Election, demonstrating the applicability of social media data in different political contexts. Metaxas et al. (2011) explored the use of Twitter activity as a predictor of electoral success, analyzing the correlation between tweet volumes and election results. Chung and Mustafaraj (2011) applied sentiment analysis to predict the 2010 US Congressional elections, underscoring the potential of Twitter data for real-time election forecasting. Effing et al. (2011) explored the role of social media in the 2010 Dutch municipal elections, highlighting the influence of online engagement on electoral outcomes. Skoric et al. (2012) demonstrated the predictive power of Twitter data in the 2011 Singapore General Election, providing evidence for the global applicability of social media-based predictions.

Burnap et al. (2016) significantly advanced the field by employing sentiment analysis of Twitter data to predict the 2015 UK General Election, demonstrating the potential of combining social media data with traditional methods.

CONTACT Dhrubajyoti Ghosh  dg302@duke.edu  Department of Biostatistics and Bioinformatics, Duke University, Durham, NC, USA

© 2025 The Author(s). Published with license by Taylor & Francis Group, LLC

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Tsakalidis et al. (2015) integrated Twitter data with traditional polling to predict multiple elections, underscoring the value of hybrid approaches. Ceron et al. (2014) proposed a supervised machine learning approach to predict the 2013 Italian parliamentary elections using Twitter data, revealing the advantages of combining sentiment analysis with volume metrics. Jungherr (2016) combined sentiment analysis and polling data to predict the 2013 German federal election, highlighting the benefits of integrating diverse data sources. Budiharto and Meiliana (2018) used sentiment analysis and network features to predict the Indonesian presidential election while Jain and Kumar (2017) applied machine learning techniques to Twitter data in India, leveraging features like tweet frequency, sentiment scores, and engagement metrics to forecast election outcomes. Beauchamp (2017) used sentiment analysis for the 2016 US Presidential primaries, showcasing the real-time predictive capabilities of social media data. Jahanbakhsh and Moon (2014) developed a model incorporating temporal dynamics and network features to predict the 2016 US Presidential Election, showing that social media activity could capture real-time shifts in voter preferences. Related work predicting national and international election outcomes using social media data include: (i) Gaurav et al. (2013) forecasting the 2012 Venezuelan presidential election, (ii) Caldarelli et al. (2014) forecasting the 2013 Italian elections, (iii) Bollen et al. (2011) predicting the 2010 US Congressional elections, (iv) DiGrazia et al. (2013) forecasting the 2012 US Presidential Election, and (v) Cram et al. (2017) predicting the 2017 UK General Election, among others, thereby further validating the utility of social media data in electoral predictions.

On the other hand, the second school of thought argues against the reliability of Twitter data for election prediction, emphasizing inherent limitations and biases. Gayo-Avello (2012) critically examined the potential of Twitter for predicting elections, highlighting significant methodological concerns. Metaxas et al. (2011) emphasized the limitations of simplistic tweet volume counts, calling for more nuanced methodologies. Gayo-Avello (2013) provided a comprehensive review, arguing that while social media offers valuable insights, it should not replace traditional polling methods due to representativeness issues and noise. Jungherr et al. (2012) critically analyzed the predictive power of Twitter data for the 2009 German federal election, finding significant discrepancies between online activity and actual results. O'Connor et al. (2010) noted discrepancies between social media sentiment and polling data in the 2008 US Presidential Election. Fink et al. (2013) analyzed the 2011 Nigerian presidential elections, concluding that Twitter data alone was insufficient for reliable predictions. Mislove et al. (2011) highlighted the demographic biases in Twitter data, affecting its predictive accuracy. Jungherr (2016) provided a critical review of Twitter's role in election forecasting, identifying significant methodological flaws. Ruths and Pfeffer (2014) discussed the limitations of social media data for studying human behavior, relevant to election forecasting.

Jungherr (2014) argued that selective exposure and echo chamber effects on social media platforms limit Twitter

data's representativeness, questioning its reliability for predictions. Lazer et al. (2014) highlighted issues such as data quality and ethical considerations in big data analytics, urging rigorous methodologies. Dubois and Blank (2018) examined bots' role in amplifying political messages on Twitter, distorting public sentiment. Barberá et al. (2016) questioned the validity of social media-based predictions in the 2012 US Presidential Election, emphasizing demographic biases. Moreno et al. (2013) critiqued the ethical implications of using social media data for research. Boyd and Crawford (2012) raised concerns about privacy and consent in social media research. Hargittai (2015) discussed the limitations of social media data's representativeness. Tufekci (2017) explored the implications of algorithmic biases in social media platforms. Brito et al. (2024) analyzed the limitations of Twitter data for political predictions. Gonzales-Bailon et al. (2014) highlighted the network biases in social media data, affecting its predictive power. Howard and Kollanyi (2016) studied the role of automated accounts in the 2016 Brexit referendum, questioning the reliability of Twitter data. These critiques highlight significant gaps and limitations in the literature, suggesting that while social media data offers valuable insights, it should be used cautiously and in conjunction with traditional methods for more accurate and reliable election predictions.

The utilization of social media for election forecasting holds significant potential from a government perspective, offering benefits that enhance democratic processes and improve policy responsiveness. Traditional data collection methods like surveys and polls often lag behind the dynamic nature of public opinion shifts during election campaigns. In contrast, social media platforms provide a continuous stream of real-time data, enabling governments to gauge public sentiment instantly and adjust strategies accordingly. This capability is crucial during elections, where timely and accurate information can inform strategic decisions and policy adjustments. Social media data can reveal immediate reactions to political events, campaign messages, and policy announcements, allowing for a more agile and responsive approach to electioneering. Moreover, the granularity of social media data allows for segmentation by demographic factors such as age, gender, and location, providing a nuanced understanding of different voter groups. This granularity helps tailor campaign messages to specific audiences, ensuring relevant and resonant communication as pointed out in Tufekci (2014). For example, messages can be customized to address the specific concerns of younger voters, urban residents, or particular ethnic groups, thereby improving the effectiveness of political communication. Additionally, social media analytics can identify key influencers within these demographics who can amplify campaign messages through their networks. The global reach of social media platforms enables governments to monitor international public opinion, which is essential for informed foreign policy decisions (cf. Howard and Kollanyi 2016). By analyzing social media trends and sentiments globally, governments can gauge the international reaction to their policies and initiatives, allowing for more informed and effective

foreign policy decisions. This capability is particularly important in an interconnected world where domestic policies often have international repercussions. Monitoring international social media trends can also provide early warnings of potential geopolitical developments, helping governments anticipate and respond to global events.

Furthermore, social media facilitates direct interaction between government officials, political candidates, and the public, thereby enhancing public engagement and transparency. This interaction not only allows citizens to voice their opinions and ask questions but also enables governments to understand the electorate's concerns and priorities more deeply, leading to more targeted and effective campaign strategies; see, for example, Effing et al. (2011) and Stieglitz and Dang-Xuan (2013). By engaging with voters on platforms they use daily, politicians can build more personal connections, potentially increasing voter turnout and political participation. The transparency afforded by social media interactions can also build trust between the government and the public, as citizens see their concerns being acknowledged and addressed in real time. Additionally, using social media data for predictive analytics can forecast election outcomes and identify emerging trends, allowing governments to address potential challenges proactively. Advanced data science techniques such as machine learning and natural language processing can analyze patterns and trends in social media activity, offering insights that traditional methods might miss (cf. Burnap et al. 2016). Predictive analytics can be particularly valuable in closely contested elections, where small shifts in public sentiment can have a significant impact on the results. Moreover, the transparency and public nature of social media discussions promote political accountability, as politicians and policies are subjected to greater scrutiny. By integrating social media insights into election forecasting, governments can enhance campaign efficiency, combat misinformation, and ultimately ensure a more responsive and effective governance system, as noted in Boyd and Crawford (2012) and Lazer et al. (2014).

Our work considers the application of statistical analysis using Twitter data combined with results of opinion polls to predict the outcomes of recent elections in Ireland and the United States. For the first analysis, this paper will aim at forecasting the final outcome of the voting for the 36th Amendment of the Irish Constitution referendum of May 25, 2018, using a model built from Twitter data and several opinion polls conducted during that period. The second analysis involves the forecasting of the electoral outcomes of the 2018 mid-term election in the United States, again using only Twitter Data and opinion polls conducted during that period. The primary differences in our attempt from earlier works, are (i) to combine the opinion polls conducted during the campaigns with the Twitter data, and (ii) to incorporate the dynamic network structure in our time series model. For (i), we note that opinion polls, representing the traditional source of information, suffer from time lags in tracking the effects of influence campaigns in quickly evolving scenarios and are often less sensitive in predicting electoral outcomes. On the other end, social media data, such as

Twitter posts, lack fair representation of the entire voting population and may not be very effective by themselves for prediction purposes. However, we show that by cleverly combining the two sources of information, it is possible to generate very accurate predictions of the election outcome, even in cases where the races are very close.

To accomplish this, we build suitable statistical models based on some important network features of the Twitter data. For campaigns where the margins of wins are reasonably large, we show that predictions based on a simple model referred to as the THOS (Twitter Hashtag based Opinion Survey) predictions are adequate in predicting the campaign outcomes. However, the simple model is not effective for predicting the outcomes when the margin of win is small. For such cases, we develop a more complex model and the corresponding predictor, referred to as the THANOS (Twitter Hashtag and Network-based Opinion Survey) model/predictor, incorporating additional network features to yield better prediction accuracy. An important contribution of the paper is to construct relevant feature variables, including the one that makes use of the Harmonic centrality measures of the associated networks. Applied to the 2018 US Elections, the THANOS model correctly predicted the winning candidates in 11 out of the top 12 closest Senate races. The remaining one (Florida) was the closest race where the margin of the winning candidate was only 0.12% and the THANOS prediction was “undecided.” In particular, the predictions generated by the THANOS model compare favorably with competing predictive modeling approaches including those of FiveThirtyEight.com.

The rest of the paper is organized as follows. [Section 2](#) describes the data sets used in the paper for predicting election campaign outcomes and it also describes relevant background information on Network analysis used in our modeling approach. [Section 3](#) develops the two predictive models, the THOS and the THANOS, combining opinion polls with Twitter data. [Sections 4](#) and [5](#) respectively describe predictions from our models for the Irish referendum on the 36th constitutional amendment and the 2018 US Senate elections, respectively. [Section 6](#) gives some discussions of the results and some concluding remarks.

2. Background

2.1. Descriptions of Data Examples

For illustrating the predictive modeling approaches of the paper, we considered two recent electoral campaigns where the results of several opinion polls and social media data were readily available. The first of these is the Ireland Vote on the 36th constitution referendum. Abortion has been subjected to criminal penalty in Ireland by the Offences against the Person Act since 1861. The Eighth Amendment of the Constitution Act 1983 inserted a subsection recognizing the equal right to life of the pregnant woman and the unborn. This was instigated by the Pro-Life Amendment Campaign for fear that the 1861 prohibition might be weakened by liberal legislators. In 1992, the X case ruled that abortion is permitted where pregnancy threatens a woman's life,

including by risk of suicide. However, no regulatory framework existed till 2013, until the introduction of the Protection of Life During Pregnancy Act 2013, which defined the circumstances and processes within which abortion in Ireland could be legally performed. This was driven by the two famous events: The A, B and C vs Ireland case in the European Court of Human Rights in 2010, and the death of Savita Halappanavar after miscarriage in 2012. The 1983 referendum is commonly viewed to equate the life of a pregnant woman with the fetus, making abortion unavailable in almost all circumstances. It is also seen as affecting maternal healthcare because a woman loses her right to refuse consent to medical treatment in pregnancy. Even though the Protection of Life During Pregnancy Act 2013 defined the circumstances under which abortion could be legally performed, it still prohibited legal abortion in many cases. The proposed 36th amendment, also known as the repeal of the 8th amendment, allows the government to legislate on abortion. The proposed legislation brought Ireland in line with the majority of European countries, allowing for abortion on request up to the 12th week of pregnancy (subject to medical regulation).

The dataset for this example consists of Twitter data collected during the campaign and opinion polls conducted by various organizations over the period December 2017 to May 2018, as shown in Table 1. We will combine the two data sources to predict the outcome of the referendum using the (simpler) THOS model, to be described in Section 3. In this case, the actual proportion of votes in favor of the amendment was 66.4%.

The second example that we consider here involves the outcome of the 2018 US Election. The election was conducted on November 6, 2018 for 35 of the 100 seats in the US Senates and all 435 seats in the US House of Representatives. In this paper, we restrict attention to the top 12 closest senate seat races where the election outcome prediction is more challenging. Table 2 gives the list of states/senate seats, the winning party, and the margins of win for the top 12 closest races.

We used opinion polls from several US organizations for predictive modeling, and the Twitter Data in this study was obtained from Wrubel et al. (2019). The Hydrator (Documenting the Now 2020) was used for hydrating the

tweets, the hydration having been conducted in April 2019. The hydrated data consisted of 1, 387, 688 tweets from 708, 916 users, over the time period of November 20, 2017 to March 1, 2019. It is to be noted here that a significant proportion of data was lost due to the tweets or the tweet account being deleted. As a safeguard, we have constructed our Twitter data network model carefully and used network features that are robust to such deletion. Indeed, to assess the effect of the bias due to this deletion process on the final model prediction, we have also conducted explicit sensitivity analysis with subsamples of the data by deleting various proportions of the available data values (ranging from 0% – 30%) and we have found almost no change in our prediction performances (cf. Section 5.5 below).

2.2. Construction of a Network for Twitter Data

Social media has emerged as a potential platform for political campaigns to gain momentum. Propaganda from political campaigns can reach a vast audience, primarily owing to mentions and retweets on Twitter. For the purpose of this paper, we have created the network by assigning a directed edge between any two users based on retweets, i.e. if user A retweeted user B, then a directed edge is constructed from B to A. Here it is to be noted that even if a certain user, say “A,” is not in the given tweet-id sample, the user is added to the network if another user “B” from the database has ever retweeted or mentioned “A” in their tweets. Hence, it is very likely that the top influencers are always included in

Table 2. Top 12 close races in the US 2018 election.

District	Winner	Margin (%)
Florida	Republican (flip)	0.12
Arizona	Democratic (flip)	2.34
Texas	Republican	2.57
West Virginia	Democratic	3.31
Montana	Democratic	3.55
Nevada	Democratic (flip)	5.03
Missouri	Republican (flip)	5.81
Indiana	Republican (flip)	5.89
Michigan	Democratic	6.51
Ohio	Democratic	6.85
Mississippi (Special)	Republican	7.27
California	Democratic	8.33

Table 1. Opinion polls for Ireland voting.

Dates conducted	Polling organisation/client	Sample size	Yes (%)	No (%)	Undecided (%)
10–15 May 2018	Red C/Sunday Business Post	1015	56	27	17
14–15 May 2018	Ipsos MRBI/Irish Times	1200	44	32	24
8–15 May 2018	Behaviour & Attitudes/The Sunday Times	935	52	34	19
30 Apr–8 May 2018	Millward Brown/Sunday Independent	1000	45	34	18
19–25 Apr 2018	Red C/Sunday Business Post	1000	53	26	20
5–17 Apr 2018	Behaviour & Attitudes/ The Sunday Times	928	47	28	26
16–17 Apr 2018	Ipsos MRBI/Irish Times	1200	47	28	25
5–12 Apr 2018	Behaviour & Attitudes/ The Sunday Times	926	49	31	21
26 Mar–3 Apr 2018	Red C/Sunday Business Post	1000	56	26	18
16–23 Mar 2018	Behaviour & Attitudes/ The Sunday Times	926	49	30	21
6–13 Feb 2018	Behaviour & Attitudes/ The Sunday Times	926	49	30	21
18–25 Jan 2018	Red C/Sunday Business Post	N/A	56	29	15
4–5 Dec 2017	Ipsos MRBI/Irish Times	1200	62	26	13

the network, even if they are removed at first due to sampling. A toy example of our network formation is given in Figure 1. As we can see from the toy example, even if we remove an influencer node (A), it will come back to the network due to the retweets done by other users. However, if we remove a fringe user (Z), it won't come back to the network since nobody has retweeted them. In our model, we are mainly concerned with the main influencers and the top hashtags, so the fringe users would not have significant impact on the network, making the network construction robust to such sampling variations.

2.3. Extraction of Network Features

To gain a better understanding of network features that are important indicators of how influence propagates through social media, the growth of the underlying network is of primary interest. Figure 2 exhibits a subset of the network structure developing over time for one particular case (Ireland 2018 Campaign). The network is created using retweets, and with four “influencers” (Users with top retweets), two from each campaign (“yes” and “no”). The nodes in the graph represent the users in the data, with a directed edge existing from node one node to another through *retweet* or *mention*. The “Influencers” form the central nodes in the four clusters (or cliques), A_1 and A_2 from the “yes” campaign, and B_1 and B_2 from the “no” campaign, all four colored red in Figure 2. These four nodes are used to take a subsample from the entire network, i.e. we only consider the retweets and mentions involving these 4 nodes, and we observe how the network grows over time, as depicted in Figure 2. As we can see in Figure 2, the network initially started with four clusters and dynamically expanded with interactions among all four clusters, the snapshots of Day 1, 10, 30 and 45 given in the figure. The network initially consists of four small, distinct clusters around these influencers on Day 1. Over time, engagement increases, and the network expands as more users interact through

retweets. By Day 10, the clusters have grown denser, with a larger number of users engaging. By Day 30, the network exhibits significant expansion, with some retweet interactions forming bridges between clusters, indicating cross-group engagement. By Day 45, the structure becomes highly complex, reflecting the widespread diffusion of information. While this subset does not represent the full campaign network, it effectively illustrates the role of high-impact users in amplifying electoral messaging. The same methodology is applied to construct the network for the U.S. campaign, allowing us to compare the dynamics of information propagation across different electoral contexts.

The growth pattern of the network in the bottom panel of Figure 2 suggests copious interactions within and among the four cliques, which indicates that the common audience are influenced by the influencers from both campaigns. The next important question that arises from this discussion is therefore how much a common user is influenced by an “influencer,” and what is the social significance of an “influencer,” i.e. the extent to which they are able to influence a campaign. A well-studied approach is to use centrality measures to define the amount of influence a user has on a network. A centrality measure can be defined as a function which assigns a score to a user, indicating the amount of interactions the user has with other nodes in the network. For example, the four influencers in Figure 2 will have high centrality measure, while those on the fringes will have low scores. We can also obtain a measure of the centrality score of the network, using graph centrality score (Freeman et al. (2002)), which calculates the interaction level of a graph based on the centrality scores of its nodes. This might indicate campaign effectiveness, or enable classification between campaign and non-campaign Twitter datasets. Here we will consider two popular measures of centrality scores:

- **Degree Centrality Score:** The Degree Centrality Score of a given node v , denoted by $C_D(v)$, is the number of edges incident upon v . It measures the first level interaction

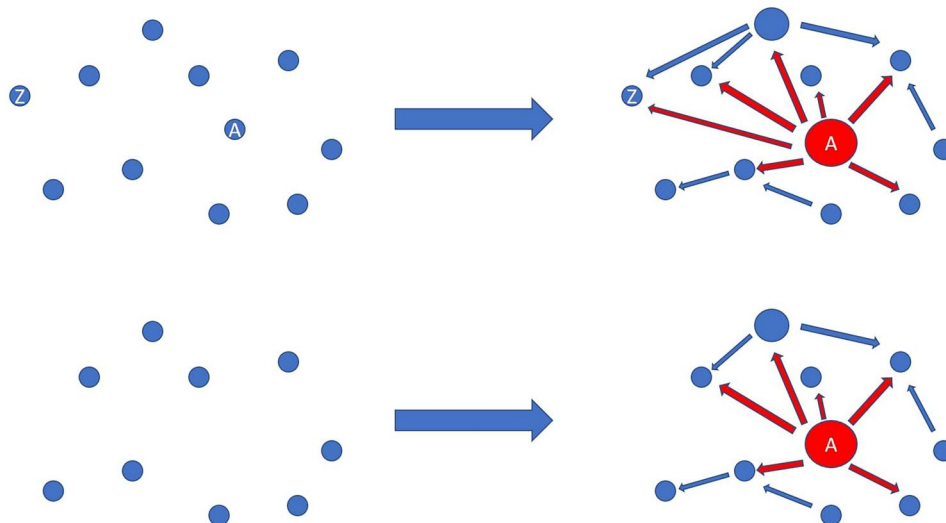


Figure 1. A toy example of the network construction in two scenarios. The upper half gives the full sample of nodes and the resulting network; the bottom half shows the resulting network on the right when node A (influencer) and node Z (fringe user) are deleted. The two networks have similar values of the network features (e.g. centrality scores of influencers) that are used in the construction of the network based model below.

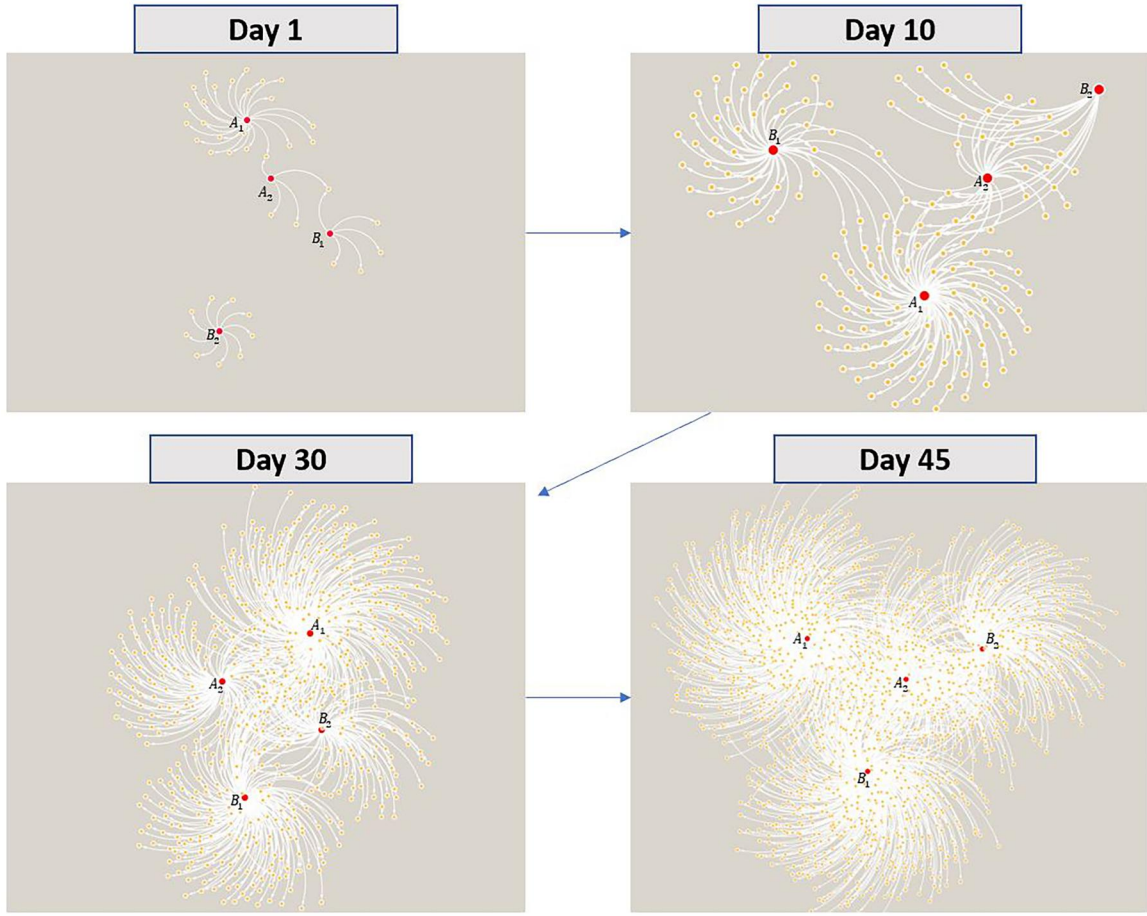


Figure 2. Network Growth in Ireland Dataset over the course of time. Here we build a subset of the network using top two influencers from both groups. As we can see the network first begins around four influential nodes and gradually expands with time, with more and more people retweeting the influencers.

experienced by user v . The graph centrality in this case is given by

$$\frac{\sum_{i=1}^{|V|} \{C_D(v^*) - C_D(v_i)\}}{|V|^2 - 3|V| + 2},$$

Where $|V|$ is the size of the Network and v^* is the node with the highest degree centrality score.

- *Harmonic Centrality* ($H(v)$) (Marchiori and Latora 2000): Let $d(y, v)$ denotes the distance between nodes y and v . The Harmonic Centrality of a node v is defined as $H(v) = \sum_{y \neq v} \frac{1}{d(y, v)}$,

In the next section, we will make use of the (dynamic) network features to build our statistical models.

3. The Models

The statistical models developed in this paper are time series regression models based on appropriately chosen features from the Twitter database. The dependent variable (y_t) is taken to be a transformation of the outcome from the opinion polls conducted during the period of the campaign. For our modeling purpose, we will only use a two-party system, which is a reasonable assumption for both of our examples. For the Ireland case, there were only two parties under

consideration while, in the US election, the Democrats and the Republicans are the only two major competitors of the elections, with the other political parties accounting for negligible vote shares. Our first model is given by:

$$\log \frac{y_t}{1 - y_t} = \beta_0 + \beta_1 \bar{x}_{t,h,l}^{(1)} + \beta_2 \bar{x}_{t,h,l}^{(2)} + \beta_3 \frac{\bar{x}_{t,h,l}^{(2)}}{\bar{x}_{t,h,l}^{(1)}} + \epsilon_t, \quad (3.1)$$

Where

- y_t := proportion of Votes obtained by Party 1 in the opinion polls,
- $\bar{x}_{t,h,l}^{(k)}$:= proportion of 10 most popular Hashtags corresponding to Party k averaged over the period of $(t - h, t - l)$, $k = 1, 2$, where $h \geq l$ determines the lagged time window,
- β_0 represents the intercept term, and $\beta_1, \beta_2, \beta_3$ coefficients corresponding to the respective predictors in the model, and
- ϵ_t := error variable at time t .

In this model, we use hashtag frequency to define the covariates. The underlying idea is that tweets coming from users leaning towards a particular party will contain more hashtags related to the corresponding party. The choice of the top 10 hashtags is arbitrary and can be tuned to get better predictions. The third feature $\frac{\bar{x}_{t,h,l}^{(2)}}{\bar{x}_{t,h,l}^{(1)}}$ gives the relative proportion of Party 2 hashtags with respect to Party 1 hashtags

that occur in the campaign dataset. We avoided delving deeper into the texts of the tweets, primarily owing to the lack of reliable sentiment analysis dictionaries in languages other than English, which is not always the most dominant language of the datasets. However, if we just consider the frequency of the most popular hashtags, we can efficiently avoid the language barrier while obtaining information from the entire dataset.

The logit transform is applied to the poll predictions to convert the support of the dependent variables to the entire real line. The most popular hashtags are classified into two groups, from either campaign and are both used in the model, along with their ratio. The parameter h signifies how much of the past history we need in order to accurately predict the final outcome and l determines how far ahead we can predict the outcome. We can provide different predictions based on different choices of h and l . Suppose that the prediction obtained from (3.1) for one choice of h and l in Model 1 is defined as $\hat{p}_{h,l}^{TH}$, where *TH* stands for Twitter Hashtag. Using this model, we can propose a prediction of the final outcome, which will be henceforth called the THOS (Twitter Hashtag based Opinion Survey) forecast, defined as

$$\bar{p}_{h,l}^{TH} = \frac{1}{|(\mathbf{h}, \mathbf{l})|} \sum_{h,l} \hat{p}_{h,l}^{TH}.$$

Thus, the final forecast $\bar{p}_{h,l}^{TH}$ is the average of all the predictions obtained by using different choices of h and l . The averaging step makes the predictor stable and guards against the effects of large and sudden changes in the polling numbers at isolated time points on the final prediction.

The above model seems to work well for elections with wide victory margins, such as the Ireland 36th amendment vote on legalizing abortion and the election for the senate seat in Connecticut in the US 2018 mid-term elections. In both of these cases, the winning party got more than 60% of votes. However, for closely fought elections (e.g. the Arizona senate seat in the 2018 US mid-term election), the THOS forecast seemed inadequate. Hence, we introduced additional network features of the Twitter data into our model. We used the number of retweets of the top influencer from either campaign (based on degree centrality) and also the harmonic centrality for the most influential nodes in the Twitter dataset. The introduction of centrality scores seemed to drastically improve our model predictions, and also the adjusted R^2 of the model. The modified model is given by:

$$\log \frac{y_t}{1 - y_t} = \beta_0 + \beta_1 \bar{x}_{t,h,l}^{(1)} + \beta_2 \bar{x}_{t,h,l}^{(2)} + \beta_3 \frac{\bar{x}_{t,h,l}^{(2)}}{\bar{x}_{t,h,l}^{(1)}} + \beta_4 h_t + \beta_5 r_t^{(1)} + \beta_6 r_t^{(2)} + \epsilon_t. \quad (3.2)$$

Here, y_t , $\bar{x}_{t,h,l}^{(k)}$ and ϵ_t are as in (3.1), with $\beta_1 - \beta_6$ being the coefficients corresponding to the predictors (such as hashtag frequencies, network centrality measures, and interactions between these variables), and the other quantities are defined as follows:

- h_t : Harmonic centrality score of the most influential user in the network at time t (for example, for the 2018 US election, it was @realDonaldTrump);

- $r_t^{(k)}$: Retweet Proportion of the most influential user from Party k at time t , $k = 1, 2$ (for example, for the 2018 US election, @realDonaldTrump for Republicans and @krassenstein for Democrats).

The aim of the second model is to improve upon the first with the use of additional network features. To provide some insight, we now discuss consideration and motivation behind the specific choices of features made in formulating model (3.2). Since the edges in the network based on the Twitter data are formed by retweets and mentions, it readily gives us an overview of the election campaign activities of the supporters of the two parties. It was observed that the top “influencers” from the two parties formed a clique around them, the size of which dynamically increased over time. The *proportion* at which (the size of) a clique around an influencer from Party 1 increases with respect to that of an influencer from Party 2 would give an indication of the public sentiment during the course of the campaign. In this project, we considered a scaled degree centrality score (based only on 1-step retweets) of the most influential user from either campaign. Additionally, we also took the Harmonic Centrality Score of the most retweeted “influencer” of the entire dataset, which is expected to capture effectiveness of the lead influencer in influencing the entire electorate (including the undecided voters). The time series using these network features, added to the earlier hashtag features, is fitted against the time series of opinion polls, to create the THANOS (Twitter Hashtag and Network-based Opinion Survey) model, given in (3.2). An important thing to be noted here is that since we are only including the network information of the top two influencers from each campaign, the procedure is very robust to sampling bias, since as already discussed, even if the influencers are not present in the initial sample, they are almost surely added to the network nodes if at least one of the users in the network had retweeted or mentioned them in their tweets.

Next, suppose that the prediction obtained from this model is defined as $\hat{p}_{h,l}^{THN}$, where we use the superscript THN (a shorthand for Twitter Hashtags and Network) to refer to the THANOS model. As in the case of the THOS model, we can provide a prediction for the probability of winning the election (by Party 1) using $\hat{p}_{h,l}^{THN}$ -s for different h, l , as

$$\bar{p}_{h,l}^{THN} = \frac{1}{|(\mathbf{h}, \mathbf{l})|} \sum_{h,l} \hat{p}_{h,l}^{THN}.$$

In the next two sections, we will apply the models to the Ireland and US elections data, and consider their accuracy in well-separated and closely fought races. We also compare our results with existing predictions put forward by some well-known election prediction websites.

4. Ireland Vote on the 36th Constitution Referendum

4.1. Data Description

In this section, we combine opinion polls and Twitter data to predict the outcome of Ireland’s vote on the Abortion

Rights amendment. Opinion polls conducted by different organizations over the period of December 2017 to May 2018 are summarized in Table 1. The sample sizes ranged from 900 to 1200, with one opinion poll having no sample size information. However, for all the opinion polls, the date and the percentage of “yes”/“no” votes were recorded. The Twitter data was collected for one and a half months, from April 13, 2018, to May 25, 2018, during the campaign for the 36th amendment (Source: Littman 2018), and the vote was conducted on May 25, 2018. The dataset contained 2,279,396 tweet ids, which was reduced to 1,933,397 tweets after passing through the Hydrator to obtain the actual tweets; It was not possible to collect tweets from users who have deleted their Twitter accounts after the collection of the tweet ids.

The time series of the times of tweets aggregated over a 3-hour window, is given in the left panel of Figure 3. The plot shows a sharp peak around May 25, the day of the vote, and a couple of small peaks during the campaign. One such peak is seen around May 13, after Facebook and Google disclosed that pro-life agencies from outside Ireland were trying to influence the campaign. Additionally, the time series of the tweets exhibited a periodic pattern. A periodogram analysis (cf. the right panel of Figure 3) reveals a sharp peak around a frequency of 0.125, giving the period to be 8. Since we have taken a granularity of 3 h, suggests a periodic pattern every 24 h.

4.2. Prediction of Outcome

For prediction of the outcome of the May 25 vote we extracted information only from the hashtags of the Twitter data. Although it might be tempting to use the actual text of the tweets, it is extremely difficult to conduct meaningful sentiment analysis. This is primarily due to the fact that even though we can take bigrams and trigrams (i.e. two- and three-word patterns) from the texts, it still might not give a clear indication as to which side the user is leaning towards. Also, as indicated earlier, the absence of a “good” sentiment dictionary covering different languages used in the tweets makes it quite difficult to use the text information. Even though majority of tweets in this particular

dataset were made in English (approximately 93%), there were several other languages used in the twitter dataset; Spanish, French and Italian being the other commonly occurring languages. As a result, it is difficult to use standard sentiment analysis tools for this case. By suitably choosing the hashtag information alone, we could generate good predictions. An advantage of this approach is that the proposed methodology can be applied more widely even in situations where the tweets are not mostly in English.

To construct relevant network features using the hashtags, we collected the most frequently occurring hashtags for the two campaigns and extracted the number of times they occurred over time in the Twitter dataset. The set of hashtags used is given in Table 3 – with the “yes” hashtags being associated with the “vote yes” campaign and “no” hashtags with the other campaign.

Note that the referendum was in effect a choice between repeal (“yes” campaign) and continuation (“no” campaign) of the provisions of the 8th Amendment of the Constitution of Ireland which explains the hashtags containing “8th” in Table 3.

Figure 4 shows the time series of the number of hashtags for the two campaigns and also the proportion of tweets in support of the “yes” campaign in 3-hour intervals over the campaign period. Although the number of hashtags for the “yes” campaign was overwhelmingly larger for the “yes” campaign than the “no” campaign, there was an indication of “mood swing” in the Twitter database. The right panel shows that there was a sharp decrease in the proportion of “yes” hashtags for the initial time period, which then increased again from around the end of April. This is also seen in the proportion of supporters of the “yes” campaign in the opinion polls. Thus, it seems logical to build a model

Table 3. Hashtags Chosen for the prediction model.

“yes” Hashtags	“no” Hashtags
<i>repealthe8th</i>	<i>savethe8th</i>
<i>togetherforyes</i>	<i>lovebothvoteno</i>
<i>voteyes</i>	<i>voteno</i>
<i>repeal</i>	<i>loveboth</i>
<i>repealedthe8th</i>	<i>votenoabortion</i>
<i>yes</i>	<i>prolife</i>
<i>together4yes</i>	<i>no</i>

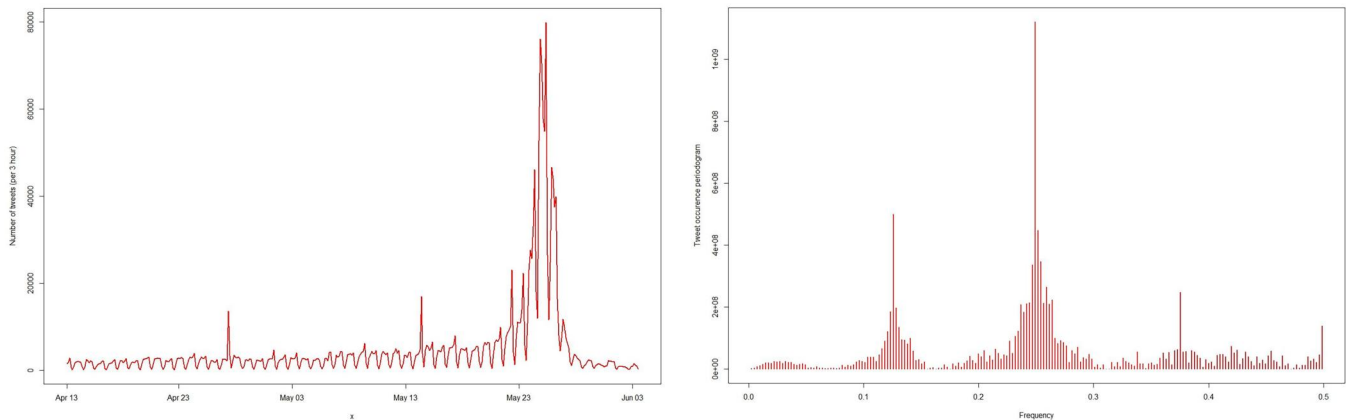


Figure 3. Time Series of Tweets. The first plot indicates the time distribution of the tweets with a granulation level of 3 h. The second plot is a periodogram of the differenced time series, indicating a peak at a frequency of 8, and thereby exhibiting the daily periodic pattern of the tweets.

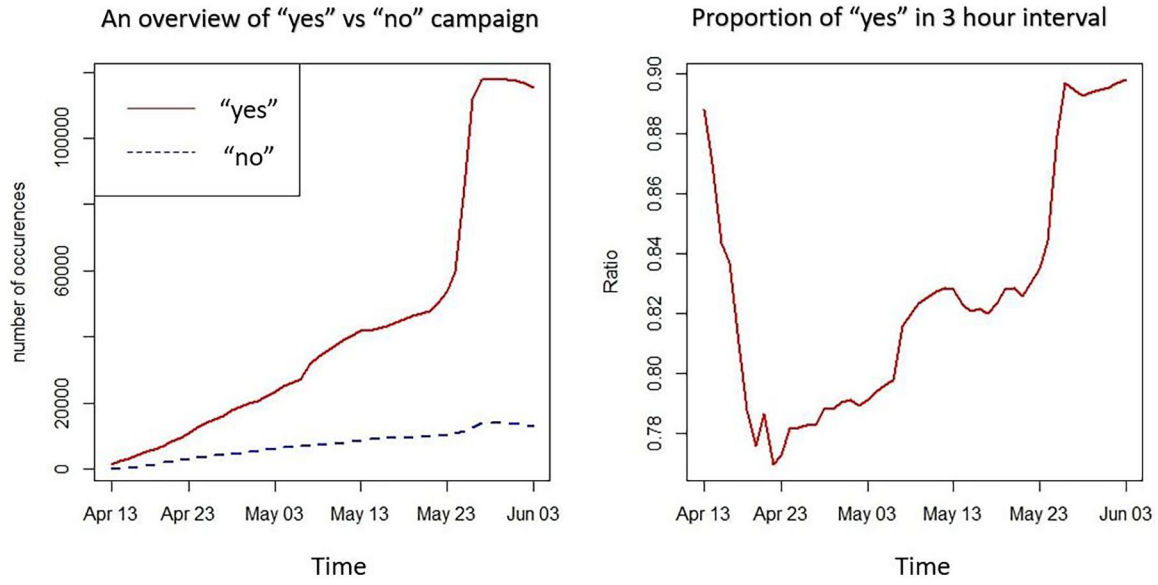


Figure 4. Hashtag Frequency of Two Parties.

using this feature, i.e. the proportion of hashtags for the “yes” campaign in the Twitter database.

In an exploratory data analysis, the THOS model was fitted for different time lags, ranging from the past 3 h to the entire history, i.e. from the day the data was available. The adjusted R^2 can be seen to flatten out after about 15 days. Thus, it was enough to consider the Twitter data for the past 15 days, since we wouldn’t lose any information if we ignored the data before that. In other words, to predict today’s vote’s outcome using Twitter, it is enough to look at the data for the last 15 days. The fit summary gives a high positive β_2 which implies that the proportion of “yes” hashtags is an important feature. The predicted THOS proportion of the final vote using this model is **0.6725**, in comparison with the actual value of **0.664**. The close prediction shows that it might actually be possible to predict the outcome of the vote using Twitter data. Even though we already got close prediction, we also applied THANOS proportion of final vote, which came out to be **0.6701**, a prediction closer to the true value.

5. US Election 2018

In this section, we will use our models to predict the outcomes of the US 2018 election using Twitter Data and opinion polling. The election was conducted on November 6, 2018, for 35 of the 100 seats in the US Senate and all 435 seats in the US House of Representatives. As indicated in Section 2, we will primarily concentrate on the senate election for our prediction purposes using the THOS and the THANOS models and compare their performance with existing predictions.

The Twitter Data in this study, obtained from Wrubel et al. (2019), consist of 1, 387, 688 tweets from 708, 916 users, over the time period of November 20, 2017 to March 1, 2019. For each senate election, we have subsampled further to select only those tweets that are relevant to the voting campaigns for that seat. In the Ireland example, the

victory margin was substantial but that was not always the case for the US election. Hence, we would need a better model for efficient predictions.

For space consideration, we will primarily focus on the results of our predictive analysis for 4 states – Connecticut (Democrats 59.53%), Montana (Democrats 50.33%), Arizona (Democrats 50%), and Florida (Republicans 50.06%). The last three of these appear in the list of top 12 close races in the US 2018 election as shown in Table 2, Section 2. Connecticut was a one-sided contest, similar to the Ireland referendum. We also report the summary of the prediction results for the remaining seats (among the top 12) in Section 5, with further details on the model performance relegated to the Appendix A. Overall, the network based THANOS model provide very accurate predictions of the true proportions for all of these relatively more difficult cases.

5.1. Connecticut

The Connecticut election experienced a clearly separated race, with the incumbent Democrat Senator Chris Murphy winning with 59.53% votes. This election outcome is similar to the Ireland votes, and we applied Model 1 to see the performance. In this case, the model can be written as:

$$\log \frac{y_t}{1 - y_t} = \beta_0 + \beta_1 \bar{x}_{t,h,l}^{(d)} + \beta_2 \bar{x}_{t,h,l}^{(r)} + \beta_3 \frac{\bar{x}_{t,h,l}^{(r)}}{\bar{x}_{t,h,l}^{(d)}} + \epsilon_t \quad (5.1)$$

Where, as in (3.1), y_t is the proportion of votes obtained by Democrats in the opinion polls and $\bar{x}_{t,h,l}^{(d)}$ and $\bar{x}_{t,h,l}^{(r)}$ are respectively the proportions of Democrat and Republican Hashtags averaged over the period of $(t-h, t-l)$. The opinion polls used in this model are from multiple sources, conducted prior to the election. The hashtags used in $\bar{x}_{t,h,l}^{(r)}$ and $\bar{x}_{t,h,l}^{(d)}$ are the top 10 hashtags from Republican and Democrat campaigns, respectively. The predictions obtained

from the model obtained from different choices of h and l are given in Table 4.

Here the prediction target is the relative percentage of Democrat votes over Republican votes, i.e. $\frac{v_d}{v_d+v_r}$. In this case, the true value was 0.6002. As we can see the predictions for all choices of h and l are pretty close to the actual value. The THOS model forecast (i.e. the average of the entries in the table) in this case is 0.5928, with relative error (defined as $|\text{prediction} - \text{true}|/\text{true}$) as 1.23%. FiveThirtyEight.com, the popular website for predictions, gave a prediction of 0.609, which is comparable to our Model 1 prediction.

5.2. Montana

In Montana, incumbent Democrat Senator Jon Tester ran against Republican candidate Matt Rosendale, keeping the seat with 50.33% votes compared to 46.78% for Rosendale. The prediction outcomes for Model 1 in this case are given in Table 5. As we can see, for many choices of h and l , we obtain correct predictions but there are some cases where the individual predictions are wrong. The THOS forecast in this case is 0.5051, with a relative error of 2.55%, the true value being 0.5183. Next, we test the performance of our second model, using the network structure of the Twitter data, to improve the prediction.

Model 2 predictions are given in Table 6 for different choices of h and l . The averaged THANOS forecast in this case turns out to be 0.5118 with a relative error of 1.25%. In comparison, the prediction given by FiveThirtyEight.com was 0.5246. Also, in the second case, almost all choices of h and l predict Democrats to be the winner in Montana, an improvement from our first model. Hence, the incorporation of network features has led to an improvement in the forecast.

Table 4. $\hat{p}^{TH}$ Forecast for Connecticut Senate in US 2018 Midterm Election for different choices of h and l .

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5953305	0.5945630	–	–	–
$h = 10$	0.5954857	0.5957060	0.5980478	–	–
$h = 15$	0.6059773	0.6077476	0.6063503	0.5646253	–
$h = 20$	0.6126172	0.6095569	0.5724295	0.5376829	–
$h = 50$	0.6133662	0.5975079	0.5889431	0.5826463	0.5912204

Table 5. $\hat{p}^{TH}$ Forecast for Montana Senate in US 2018 Midterm Election for different choices of h and l .

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5495463	0.5473994	–	–	–
$h = 10$	0.5318327	0.5268676	0.4984408	–	–
$h = 15$	0.5219245	0.5148890	0.4898930	0.4736951	–
$h = 20$	0.5085644	0.4990497	0.4594693	0.4555957	–
$h = 50$	0.4927631	0.4970696	0.510152	0.5011334	0.5128071

Table 6. $\hat{p}^{THN}$ Forecast for Montana Senate in US 2018 Midterm Election for different choices of h and l .

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5222159	0.5227301	–	–	–
$h = 10$	0.5190015	0.5172068	0.5097920	–	–
$h = 15$	0.5201558	0.5196510	0.5018776	0.5261745	–
$h = 20$	0.5195608	0.5191532	0.5019673	0.4899208	–
$h = 50$	0.4950174	0.5031124	0.5090309	0.4859780	0.5123795

5.3. Arizona

In Arizona, incumbent Republican senator Jeff Flake did not run for re-election. The Democratic candidate was Kyrsten Sinema, while the Republican candidate was Martha McSally. It was a close-fought election, with Sinema winning with 50% vote, compared to McSally's 47.61%. The predictions from Model 1 and Model 2 are given in Tables 7 and 8, respectively.

As we can see, Model (3.1) is not sensitive enough in this case, giving wrong predictions for many values of h and l . The averaged THOS forecast here is 0.4867 (true value: 0.5122), with a relative error of 4.98%. However, Model 2 gives a much better prediction, with $\bar{p}_{h,l}^{THN}$ being 0.5116, with a relative error of 0.117%. The FiveThirtyEight.com prediction in this case was 0.5087. Hence, in this case, Model 2 is performing decidedly better than Model 1. In fact, the THANOS forecast performs better than existing election prediction algorithms as shown in Figure 5.

5.4. Florida

The Florida election was the closest election in the US 2018 election, with incumbent Democratic Senator Bill Nelson (49.33% votes) narrowly defeated by Republican Candidate Rick Scott (50.06% votes). Both our models gave very close predictions to the actual outcome (0.4994). The value of $\bar{p}_{h,l}^{TH}$ for Model 1 was 0.5035 (relative error: 0.821%), while that for Model 2 was 0.5025 (relative error 0.621%). Since every prediction has some variability associated with it, it is sensible to mark a contest Neutral if the predicted value is so close to 0.5. FiveThirtyEight.com gave a close wrong prediction, with the prediction value being 0.516 (relative error: 3.32%). Hence, our prediction method is in fact out-performing in this particular scenario. Figure 5 provides a graphical illustration comparing the results of the two models with those of existing platforms, including FiveThirtyEight.com. It is evident that the THANOS forecast provides closer approximation of the final outcome, compared to existing platforms, including the FiveThirtyEight.com and PredictIt predictions.

Table 7. $\hat{p}^{TH}$ Forecast for Arizona Senate in US 2018 Midterm Election for different choices of h and l .

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.4943676	0.4932947	–	–	–
$h = 10$	0.4997875	0.4997739	0.5043522	–	–
$h = 15$	0.4962760	0.4939905	0.4847742	0.4615345	–
$h = 20$	0.4800426	0.4708316	0.4360083	0.4455570	–
$h = 50$	0.4962980	0.4934741	0.4974655	0.5107796	0.5011974

Table 8. $\hat{p}^{THN}$ Forecast for Arizona Senate in US 2018 Midterm Election for different choices of h and l .

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5048349	0.5112638	–	–	–
$h = 10$	0.5116958	0.5131329	0.5092351	–	–
$h = 15$	0.5080966	0.5109581	0.5111811	0.5246686	–
$h = 20$	0.5073115	0.5055461	0.5021870	0.5474827	–
$h = 50$	0.5033984	0.5035030	0.5150762	0.5121114	0.5071346

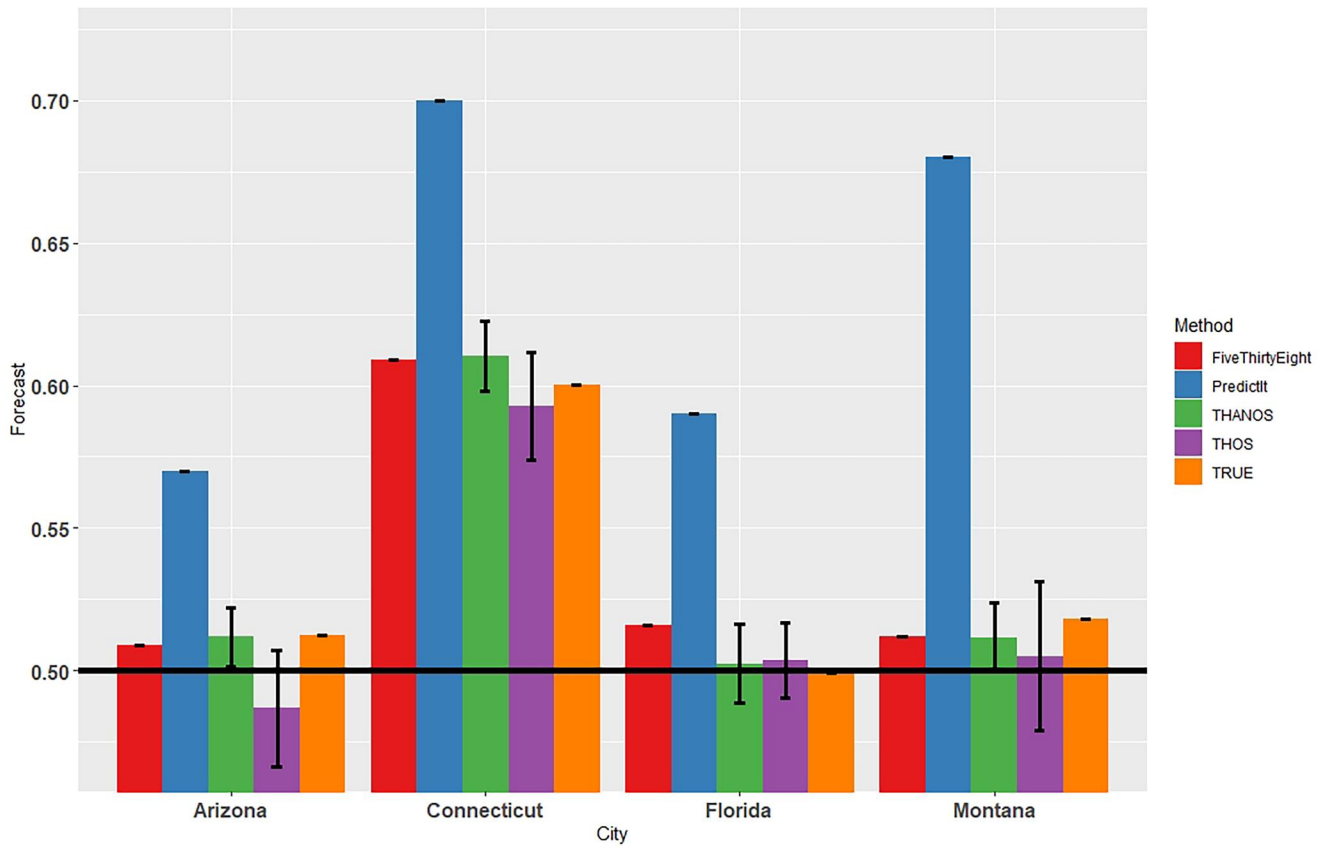


Figure 5. Comparison of the result of our model with other existing prediction platforms.

Table 9. THOS and THANOS results for the top 12 closely fought senate seats in the 2018 US Election.

Senate Seat	TRUE	THOS	THANOS
Connecticut	0.6002	0.5928	0.5947
Arizona	0.5122	0.4867	0.5116
Florida	0.4994	0.5035	0.5025
Indiana	0.5181	0.5085	0.5056
Missouri	0.5299	0.4962	0.5117
Minnesota	0.6248	0.6303	0.6286
Tennessee	0.4453	0.4725	0.4596
Pennsylvania	0.5666	0.5737	0.5606
Virginia	0.5816	0.6191	0.5841
Nevada	0.5261	0.5017	0.5027
New Mexico	0.6395	0.6018	0.6371
New York	0.6700	0.6454	0.6626

5.5. The Other States in the Group of Top 12

We further conducted our prediction on eight more states. Details of the analyses for these eight seats are given in Appendix A below and the prediction results are summarized in Table 9. As we can see, both our THOS and THANOS models performed very well, the THOS performing a little poorly in closely fought cases. However, the THANOS is giving predictions that are very close to the true winning proportion.

5.6. Model Robustness

The Twitter data used in this analysis are subject to a potential biasing effect that arises from deletion of some tweets, e.g. deletion of tweets between the collection date of the

Table 10. Sensitivity analysis using subsamples.

Seat	True	Full	10 %	15 %	20 %
Arizona	0.5122	0.5116	0.5114	0.5112	0.5104
Indiana	0.5181	0.5056	0.5056	0.5101	0.5052
Virginia	0.5816	0.5841	0.5962	0.5761	0.5608
Tennessee	0.4453	0.4596	0.4381	0.4762	0.4409

The %-figures in the top row indicate the fraction of nodes deleted.

tweet-ids and the hydration date. To assess the effect of the tweet deletion on the predictions produced by our models, we have done a subsample analysis, where we randomly removed a portion of the data and recomputed the predictions. We chose four states (Arizona, Indiana, Virginia and Tennessee) with true winning fraction in the range 0.44 – .59 and with a couple of seats having a relatively small margin of winning proportion (namely, around .51). We deleted successively 10%, 20% and 30% of the tweets, thereby giving 1, 248, 988, 1, 110, 211 and 971, 434 tweets respectively. Then we conducted the THANOS prediction; the results are reported in Table 10. It is evident from this exercise that the predictions based on the remaining subsets of the data were very close to the one we obtained using the full data for all the states. As result, the THANOS model is robust to the effect of tweet deletion to a significant proportion.

The robustness of the THANOS model can also be explained by considering the specific way we constructed the network and by considering the network features used for constructing the model. Note that much of the twitter database is populated by fringe tweets, i.e. by tweets from users tweeting only once or twice during the course of the

events. The popular hashtags, on the other hand, appear with significant frequency even after the deletion of a given fraction of the dataset and this essentially results in a new network that is very similar to the network based on the full dataset. In fact, as described in [Section 2.2](#), our network construction method is dependent mainly on the most influential users who have been retweeted most times, and that proportion of retweets remained more or less the same within the reduced dataset even after deletion of the tweets. As a result, our method is quite robust to tweet deletion, and can take care of the “deletion bias” which occurs due to deleted tweets or accounts.

5.7. Discussion

The results from our proposed models, THOS and THANOS, show that integrating Twitter data with opinion polls can provide significant improvements in forecasting electoral outcomes, especially in closely contested elections. The two models—Twitter Hashtag based Opinion Survey (THOS) and Twitter Hashtag and Network-based Opinion Survey (THANOS)—were tested using data from the 36th amendment referendum in Ireland and the 2018 US Senate elections. Our findings underscore the utility of leveraging real-time, dynamic social media data alongside traditional polling methods to enhance prediction accuracy.

In our analysis, the THOS model demonstrated its effectiveness in predicting elections with clear frontrunners. This was particularly evident in the case of the Ireland referendum on the 36th amendment, where the prediction made by THOS was very close to the actual vote outcome, with a relative error of only 1.23%. This highlights the value of Twitter data, especially in scenarios where the political landscape is less volatile, and social media activity directly correlates with public sentiment. However, the THOS model fell short in the case of closely contested elections, such as the 2018 US Senate races, where margin differences were narrower. In such cases, the THANOS model outperformed THOS by incorporating additional network features, including centrality measures that capture the influence of key users within the Twitter network. The inclusion of network structure, especially centrality scores, improved the model's ability to forecast outcomes accurately, even in the tightest races. For example, the THANOS model accurately predicted the winners in 11 out of the top 12 closest Senate races, including the contentious Arizona race, where the margin was only 2.4%.

One of the most significant contributions of our work is the demonstration that the dynamics of online networks, particularly the influence of central figures or influencers, can substantially improve electoral predictions. Our results from the THANOS model suggest that the centrality of influential users in Twitter networks is a critical determinant of electoral outcomes. By capturing the interactions of influential campaign figures and their followers, the THANOS model is able to provide deeper insights into voter behavior, which traditional polling methods might miss. This dynamic approach is particularly important as it allows for real-time

adjustments to predictions, incorporating new data as the campaign unfolds.

A key strength of our methodology is its adaptability to different electoral systems. Although we have focused on two-party systems in this paper, our approach could be extended to multi-party systems or referenda, where multiple outcomes are possible. The THOS and THANOS models are flexible enough to handle variations in electoral systems, provided that the relevant features, such as hashtag frequency and network centrality, are appropriately adjusted. This flexibility opens the door for the application of these models to a wide range of elections, from local government races to national referenda.

In terms of model robustness, our results demonstrate that the THANOS model is resilient to data deletion and network sparsity. We conducted sensitivity analyses by removing portions of the data, simulating the potential effects of tweet deletions or account removals. The results showed that the model's predictions remained relatively stable, indicating that the centrality measures of key influencers remain reliable even in the face of data loss. This robustness is crucial for ensuring the reliability of the model in real-world applications, where data may be incomplete or subject to biases from deleted accounts and tweets. Additionally, the model's reliance on network features rather than raw tweet volume further enhances its robustness, as it reduces the sensitivity to short-term fluctuations in online activity.

While our models show promise, there are notable limitations and areas for future work. One such limitation is the potential bias introduced by the demographics of Twitter users. As highlighted by previous research ([Mislove et al. 2011](#); [Jungherr 2016](#)), social media platforms like Twitter are often skewed toward younger, more urban populations. This demographic bias could affect the generalizability of our predictions, particularly in regions where Twitter penetration is lower or where demographic representation is imbalanced. For instance, in countries with low Twitter usage, such as in parts of Eastern Europe or the Middle East, the models may suffer from a lack of representative data, as the platform may not reflect the full spectrum of the voting population. This limitation underscores the need for caution when applying social media-based models in these regions. Future research could explore how to mitigate these biases by incorporating data from other social media platforms or by adjusting for demographic imbalances through weighting techniques. Additionally, while our work successfully combined Twitter data with opinion polls, it is worth noting that Twitter data alone may not always be sufficient to predict electoral outcomes accurately, especially in countries with limited Twitter usage. The addition of other data sources, such as Facebook or even SMS-based surveys, could provide a more comprehensive view of public sentiment, particularly in regions where Twitter is not as prevalent. Moreover, the volume of data from Twitter may be overwhelming in some elections, leading to issues with data sparsity or noise, particularly when dealing with minor political parties or elections with less social media activity. This presents an interesting challenge for future studies to

explore methods for balancing the inclusion of robust data without allowing noise to compromise the accuracy of predictions.

6. Concluding Remarks

In this paper, we have proposed two methods of forecasting elections, combining Twitter data and opinion polls conducted during the campaigns. The basic idea is that social media data can be used efficiently to improve upon traditional opinion polls, and can provide an accurate forecast of election results. The simple THOS forecast, where we only use the hashtag frequency of the Twitter data, proved to be effective for elections where the victory margin was wide (viz., Ireland vote, Connecticut Election). However, for close races with small victory margins, the THOS forecast seemed inadequate (e.g. Arizona). In this scenario, we proposed the THANOS forecast, which also incorporates network features of the Twitter dataset, in particular the centrality of the top influential persons. This second model proved to be very efficient in predicting elections, even when the victory margins were very small. For example, in Arizona election, the victory margin was as low as 2.4%, which the second model predicted with a relative error of only 0.117%, a prediction that proved to be even better than the widely acclaimed FiveThirtyEight.com prediction.

A key contribution of our predictive modeling approach is to identify network features that are critical for high accuracy of predictions using social media data. In particular, the network centrality features used in our analysis indicate that the users who occupy central or pivotal positions in a campaign, have significant influence and provide important clues to the final outcome of an election.

One potential concern with the use of social media data for forecasting purposes is whether the social media users provide a representative sample of the voting community. Wojcik and Hughes (2019) discusses the possibility of an inherent bias among the Twitter users towards the Democratic ideology. While the issue of presence or absence of such bias among Twitter users needs further investigation, by combining Twitter data with opinion polls, our prediction approach seems to be robust against such perceived bias and is able to provide accurate predictions of the election results that are comparable to the best available competing methods based on alternative data sources including site-specific historical election data.

There is significant scope of further research in this area. For example, finding an optimum choice of the vector $\mathbf{h}$ and $\mathbf{I}$ is of theoretical interest. One can also consider adding other features of the Twitter dataset that might improve the predictive power of the model. Some such potential choices are sentiment analysis, other centrality measures, local network features, etc. To summarize the contribution at this point, we have devised a methodology to predict election outcomes by coupling opinion polls with social media data which proves to be quite accurate in the real data applications considered in this work.

Acknowledgement

We thank the Editor and the Associate Editor for prompt handling of the manuscript and also thank a referee for a number of insightful suggestions that improved the exposition of the paper.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

This material is based upon work supported in whole or in part with funding from the Laboratory for Analytic Sciences (LAS). Any opinions, findings, conclusions, or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the LAS and/or any agency or entity of the United States Government.

ORCID

Dhrubajyoti Ghosh  <http://orcid.org/0000-0002-3360-3786>

References

- Barberá P, et al. 2016. Less is more? how demographic sample weights can improve public opinion estimates based on twitter data. *Work Pap NYU*. 37:1–37.
- Beauchamp N. 2017. Predicting and interpolating state-level polls using twitter textual data. *Am J Polit Sci*. 61(2):490–503.
- Bermingham A, Smeaton AF. 2011. On using twitter to monitor political sentiment and predict election results. *Sentiment Analysis where AI meets Psychology (SAAIP): DCU Research Repository*. 2:1–9.
- Bollen J, Mao H, Zeng X. 2011. Twitter mood predicts the stock market. *J Comput Sci*. 2(1):1–8.
- Boyd D, Crawford K. 2012. Critical questions for big data: provocations for a cultural, technological, and scholarly phenomenon. *Inform Commun Soc*. 15(5):662–679.
- Brito K, Silva Filho RLC, Adeodato PJJ. 2024. Stop trying to predict elections only with twitter – there are other data sources and technical issues to be improved. *Government Inform Q*. 41(1):101899. <https://www.sciencedirect.com/science/article/pii/S0740624X23000990>.
- Budiharto W, Meiliana M. 2018. Prediction and analysis of indonesia presidential election from twitter using sentiment analysis. *J Big Data*. 5(1):1–10.
- Burnap P, Gibson R, Sloan L, Southern R, Williams M. 2016. 140 characters to victory?: Using twitter to predict the uk 2015 general election. *Electoral Studies*. 41:230–233.
- Caldarelli G, Chessa A, Pammolli F, Pompa G, Puliga M, Riccaboni M, Riotta G. 2014. A multi-level geographical study of italian political elections from twitter data. *PLoS One*. 9(5):e95809.
- Ceron A, Curini L, Iacus SM, Porro G. 2014. Every tweet counts? how sentiment analysis of social media can improve our knowledge of citizens' political preferences with an application to italy and france. *New Media Soc*. 16(2):340–358.
- Chung J, Mustafaraj E. 2011. Can collective sentiment expressed on twitter predict political elections?. In: *Proceedings of the AAAI Conference on Artificial Intelligence*; vol. 25. p. 1770–1771.
- Cram L, Llewellyn C, Hill R, Magdy W. 2017. UK general election 2017: a twitter analysis. *arXiv preprint arXiv:170602271*.
- DiGrazia J, McKelvey K, Bollen J, Rojas F. 2013. More tweets, more votes: social media as a quantitative indicator of political behavior. *PLoS One*. 8(11):e79449.
- Documenting the Now 2020. Hydrator [computer software]. Available from: <https://github.com/docnow/hydrator>.

- Dubois E, Blank G. 2018. The echo chamber is overstated: the moderating effect of political interest and diverse media. *Information, Communication & Society*. 21(5):729–745.
- Effing R, Van Hillegersberg J, Huibers T. 2011. Social media and political participation: are facebook, twitter and youtube democratizing our political systems? In: *Electronic Participation: Third IFIP WG 8.5 International Conference, ePart 2011, Delft, The Netherlands, Aug 29–2011 Sept 1, Proceedings 3*. Springer. p. 25–35.
- Fink C, Bos N, Perrone A, Liu E, Kopecky J. 2013. Twitter, public opinion, and the 2011 nigerian presidential election. In: *2013 International Conference on Social Computing*. IEEE. p. 311–320.
- Freeman LC, et al. 2002. Centrality in social networks: conceptual clarification. *Soc Netw*. 1:238–263.
- Gaurav M, Srivastava A, Kumar A, Miller S. 2013. Leveraging candidate popularity on twitter to predict election outcome. In: *Proceedings of the 7th Workshop on Social Network Mining and Analysis*. p. 1–8.
- Gayo-Avello D. 2012. I wanted to predict elections with twitter and all i got was this lousy paper. A balanced survey on election prediction using Twitter data. *arXiv preprint arXiv:1204.6441*.
- Gayo-Avello D. 2013. A meta-analysis of state-of-the-art electoral prediction from twitter data. *Soc Sci Comp Rev*. 31(6):649–679.
- Gonzales-Bailon S, Borge-Holthoefer J, Rivero A, Moreno Y. 2014. The dynamics of protest recruitment through an online network. *Sci Rep*. 4:5571.
- Hargittai E. 2015. Is bigger always better? potential biases of big data derived from social network sites. *ANN Am Acad Polit Soc Sci*. 659(1):63–76.
- Howard PN, Kollanyi B. 2016. Bots, #strongerin, and #brexit: computational propaganda during the uk-eu referendum. *arXiv preprint arXiv:160606356*.
- Jahanbakhsh K, Moon Y. 2014. The predictive power of social media: on the predictability of us presidential elections using twitter. *arXiv preprint arXiv:14070622*.
- Jain VK, Kumar S. 2017. Towards prediction of election outcomes using social media. *IJISA*. 9(12):20–28.
- Jungherr A. 2014. The logic of political coverage on twitter: temporal dynamics and content. *J Commun*. 64(2):239–259.
- Jungherr A. 2016. Twitter use in election campaigns: a systematic literature review. *J Inform Technol Polit*. 13(1):72–91.
- Jungherr A, Jürgens P, Schoen H. 2012. Why the pirate party won the German election of 2009 or the trouble with predictions: a response to Tumasjan, a., Sprenger, to, Sander, pg, & Welpe, im “predicting elections with Twitter: what 140 characters reveal about political sentiment. *Soc Sci Comput Rev*. 30(2):229–234.
- Lazer D, Kennedy R, King G, Vespignani A. 2014. The parable of google flu: traps in big data analysis. *Science*. 343(6176):1203–1205.
- Littman J. 2018. Ireland 8th Tweet Ids. Available from DocNow.
- Marchiori M, Latora V. 2000. Harmony in the small-world. *Physica A*. 285(3-4):539–546.
- Metaxas PT, Mustafaraj E, Gayo-Avello D. 2011. How (not) to predict elections. In: *2011 IEEE Third International Conference on Privacy, Security, Risk and Trust and 2011 IEEE Third International Conference on Social Computing*. IEEE. p. 165–171.
- Mislove A, Lehmann S, Ahn YY, Onnela JP, Rosenquist J. 2011. Understanding the demographics of twitter users. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 5. p. 554–557.
- Moreno MA, Goniü N, Moreno PS, Diekema D. 2013. Ethics of social media research: common concerns and practical considerations. *Cyberpsychol Behav Soc Netw*. 16(9):708–713.
- O'Connor B, Balasubramanian R, Routledge B, Smith N. 2010. From tweets to polls: linking text sentiment to public opinion time series. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 4. p. 122–129.
- Ruths D, Pfeffer J. 2014. Social media for large studies of behavior. *Science*. 346(6213):1063–1064.
- Skoric M, Poor N, Achananuparp P, Lim EP, Jiang J. 2012. Tweets and votes: a study of the 2011 singapore general election. In: *2012 45th Hawaii International Conference on System Sciences*. IEEE. p. 2583–2591.
- Stieglitz S, Dang-Xuan L. 2013. Emotions and information diffusion in social media—sentiment of microblogs and sharing behavior. *J Manage Inf Syst*. 29(4):217–248.
- Tsakalidis A, Papadopoulos S, Cristea AI, Kompatsiaris Y. 2015. Predicting elections for multiple countries using twitter and polls. *IEEE Intell Syst*. 30(2):10–17.
- Tufekci Z. 2014. Big questions for social media big data: representativeness, validity and other methodological pitfalls. In: *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 8. p. 505–514.
- Tufekci Z. 2017. *Twitter and tear gas: the power and fragility of networked protest*. Yale University Press.
- Tumasjan A, Sprenger TO, Sandner PG, Welpe IM. 2010. Predicting elections with Twitter: what 140 characters reveal about political sentiment. *ICWSM*. 4(1):178–185.
- Wojcik S, Hughes A. 2019. Sizing up Twitter users. *PEW Res Center*. 24:1–23.
- Wrubel L, Littman J, Kerchner D. 2019. 2018 U.S. Congressional election Tweet Ids. Harvard Dataverse. Available from: <https://doi.org/10.7910/DVN/AEZPLU>.

Appendix A. Results for the other states listed in Section 5.5

Table A.1. $\hat{p}^{TH}$ Forecast for Indiana Senate in US 2018 Midterm Election for different choices of h and l Prediction: 0.5085334, True: 0.5180227.

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5018618	0.5032460	–	–	–
$h = 10$	0.5049608	0.5053055	0.5090688	–	–
$h = 15$	0.5090005	0.5078010	0.5103339	0.5132814	–
$h = 20$	0.5126853	0.5144639	0.5358764	0.5110012	–
$h = 50$	0.5178660	0.5138337	0.5057563	0.4848076	0.4924515

Table A.2. $\hat{p}^{THN}$ Forecast for Indiana Senate in US 2018 Midterm Election for different choices of h and l Prediction: 0.5056458, True: 0.5180227.

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.5137482	0.5105588	–	–	–
$h = 10$	0.5063256	0.5062263	0.5102760	–	–
$h = 15$	0.5106390	0.5102780	0.4974466	0.4812186	–
$h = 20$	0.5034794	0.5032581	0.5173822	0.4901326	–
$h = 50$	0.5156062	0.5114800	0.5156097	0.4960894	0.5018688

Table A.3. $\hat{p}^{TH}$ Forecast for Missouri Senate in US 2018 Midterm Election for different choices of h and l Prediction: 0.4962197, True: 0.5298969.

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.4958075	0.4955540	–	–	–
$h = 10$	0.4981390	0.4993187	0.4995892	–	–
$h = 15$	0.4994267	0.4993502	0.4931603	0.4934035	–
$h = 20$	0.4929507	0.4894089	0.4933019	0.4967059	–
$h = 50$	0.5025358	0.5019489	0.4980703	0.4907983	0.4924845

Table A.4. $\hat{p}^{THN}$ Forecast for Missouri Senate in US 2018 Midterm Election for different choices of h and l Prediction: 0.511716, True: 0.5298969.

	$l = 1$	$l = 2$	$l = 5$	$l = 10$	$l = 20$
$h = 5$	0.4966026	0.4973309	–	–	–
$h = 10$	0.4982224	0.4985652	0.4982907	–	–
$h = 15$	0.5031305	0.5045383	0.5076564	0.5104432	–
$h = 20$	0.5072204	0.5081681	0.5260894	0.5114477	–
$h = 50$	0.5044144	0.5039501	0.4910373	0.4784361	0.4853452